

Conservation of angular momentum in RRKM theory. A three-dimensional model for nonlinear molecules

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Abstract

The “usual” implementation of the unimolecular rate constant in RRKM theory treats nonlinear molecules as pseudo-diatomics and thus does not conserve total angular momentum. A three-dimensional model for conservation of total angular momentum for vibrator transition states with the K rotational quantum number either active or adiabatic is presented and a comparison made between this model and the “usual” one in the low-pressure limit for two different reactions. This showed that the three-dimensional model resulted in significantly larger rate constants when the K rotor was active, indicating the importance of conservation of total angular momentum.

1. Introduction

In RRKM theory, the usual approximation for conservation of angular momentum for non-linear molecules is to assume that the molecule and transition state (TS) are pseudo-diatomic moieties in which the third degree of rotational freedom (frequently called the “ K rotor”) is an independent one-dimensional rotor [1–4]. Angular momentum is then conserved for the molecule and TS as a diatomic moiety rather than the total angular momentum [1,4].

Further, it is frequently assumed that the K rotor is coupled to the internal modes in both the molecule and TS and is thus an active degree of freedom, this is particularly the case for bond breaking reactions [5], and has been suggested for isomerizations too [6]. Whether or not the K rotor is assumed active does not affect the value of the thermal high-pressure

rate constant (k_∞), but it does lead to different values for the microscopic rate constant, $k(E, J)$, and thus to different values for the thermal rate constant in the fall-off region, k_{uni} , and at the low-pressure limit (k_0).

For the purposes of this discussion, the above set of assumptions will constitute the two-dimensional model for conservation of angular momentum, which, as pointed out by Miller [7], does not conserve total angular momentum unless appropriate restrictions are put on the angular momentum of the K rotor (in addition to refs. [4,7], see also refs. [8–11]). However, Gilbert and Smith [1] have argued that the two-dimensional model should be a good approximation to the conservation of total angular momentum and that the ease of implementation that results from this approximation is worth the loss in accuracy.

Quack and Troe [8] and Miller [7] have both shown that a three-dimensional model is necessary to properly conserve angular momentum in unimolecular reaction rate theory, with the work of Miller [7] dealing specifically with RRKM theory. In these

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models, the molecule and TS are approximated as symmetric top rotors in which the two equal moments of inertia are assigned to the x and y axes and the z axis is the symmetry axis of the top. In this approximation the rotational motion of the system is determined by three quantum numbers; J , which determines the total angular momentum; M_J , which determines the component of the angular momentum along a space-fixed axis; and K , which determines the component along the symmetry, or z , axis of the system. These latter two are obviously limited by the total angular momentum, and are restricted to integer values from $-J$ to $+J$. The “appropriate restrictions” mentioned above are that the angular momentum in the z axis (K rotor) be given by

$$L_z = K\hbar, \quad K = -J \text{ to } +J.$$

The rotational energy, E_r , is the sum of the rotational kinetic energy about each of the molecule's axes, and for a symmetric top is given by

$$E_r(J, K) = \frac{[J(J+1) - K^2]\hbar^2}{2I_x} + \frac{K^2\hbar^2}{2I_z}. \quad (1)$$

Both J and M_J are “good” quantum numbers, in that they can only be changed by application of an outside torque, thus, they are adiabatic in the RRKM sense [4,7]. For a strictly rigid rotor, K would also be adiabatic, however, for real molecules it is possible that K may be an active quantum number, coupling rotational energy about the z axis with the internal modes. If so, it is still limited to values of $-J$ to $+J$ by conservation of angular momentum [7–11].

In the three-dimensional model with J and K adiabatic and all internal degrees of freedom active, the microscopic rate constant becomes a function of E , the total energy, and J , and K . Thus

$$k(E, J, K) = \frac{r^+ G_i^+(E - E_0 - E_r^+(J, K))}{h N_i(E - E_r(J, K))}, \quad (2)$$

where $G_i^+(E)$ is the sum of internal states for the TS up to the energy E , $N_i(E)$ is the density of internal states for the molecule at the energy E , E_0 is the critical energy for reaction, and r^+ is the reaction path degeneracy.

Zhu and Hase [12] and most recently, Zhu, Chen, Hase and Kaiser [13] have shown that the microcanonical average of $k(E, J, K)$ over all K states, given by

$$\begin{aligned} \langle k(E, J) \rangle_K &= \frac{r^+ \sum_K G_i^+(E - E_0 - E_r^+(J, K))}{h \sum_K N_i(E - E_r(J, K))}, \end{aligned} \quad (3)$$

is numerically identical to the microscopic rate constant obtained with K active, $k^*(J, K)$. In each case all internal degrees of freedom are active. This important result emphasizes the similarities in the microscopic rate constants for K active and adiabatic when total angular momentum is conserved. However, the macroscopic rate constants in the fall-off, k_{uni} , and at the low-pressure limits, k_0 , are different for these two cases.

In this work we will formulate a different average of $k(E, J, K)$ when K is adiabatic, one over only the reacting K states. This average will point out the differences in the microscopic rate constants for K adiabatic and active that lead to different values of k_0 . These effects will be illustrated for both a prolate and oblate top, and the results compared with those calculated using the two-dimensional model.

2. K adiabatic

We want to form the microcanonical average of $k(E, J, K)$ over reacting K states. A state will lead to reaction if the active energy in the transition state, E_a^+ , is positive. Thus, this new summation must be limited to values of J and K in both the numerator and denominator that satisfy the inequality in

$$E_a^+ = E - E_0 - E_r^+(J, K) > 0. \quad (4)$$

Let us indicate such a summation by Σ^+ , then the microcanonical average of $k(E, J, K)$ over the reacting states, $\langle k(E, J) \rangle_R$, is given by

$$\langle k(E, J) \rangle_R = \frac{r^+ \sum_K^+ G_i^+(E - E_0 - E_r^+(J, K))}{h \sum_K^+ N_i(E - E_r(J, K))}, \quad (5)$$

$$K = -J \text{ to } +J.$$

Clearly, Eq. (5) approaches Eq. (3) for large E or small J , conditions that assure that $E_a^+ > 0$ for all K , but will otherwise be larger than Eq. (3), particularly at energies near threshold. One can write an excellent approximation to the strong collision unimolecular rate constant with K adiabatic, k_{uni}^d , by substituting

$\langle k(E, J) \rangle_R$ for each value of $k(E, J, K)$ in the rate equation; this yields

$$k_{\text{uni}}^d = \int_{E_0}^{\infty} \sum_J \langle k(E, J) \rangle_R \frac{w}{w + \langle k(E, J) \rangle_R} \times \frac{(2J+1) \sum_K^+ N_i(E - E_r(J, K))}{Q_{\text{int}} Q_r} e^{-E/kT} dE. \quad (6)$$

This approximation (Eq. (6)) is exact at the low- and high-pressure limits and deviates slightly from the exact in the intermediate pressure ranges (limited calculations have indicated that this deviation is 5%–20%). In this equation, w is the collision frequency and the summation over J is limited to values for which $\langle k(E, J) \rangle_R > 0$ rather than $J(J+1)\hbar^2/2I_x < E - E_0$, as in the two-dimensional model. In the high- and low-pressure limits, Eq. (6) agrees with the adiabatic/adiabatic model of Zhu et al. [13], who obtained the exact value of k_{uni}^d by summation over both K and J .

3. K active

When the K rotor is active, the rotational energy about the z axis, which we will call E_K^* and E_K , is coupled with the internal energy in both the molecule and TS (the active/active model of ref. [13]). The energy in this rotation is $K^2\hbar^2/2I_z$ (with a similar expression for the TS), which we take to be the active rotational energy. Previously, it has been suggested that the active energy could be approximated instead by $\frac{1}{2}K^2\hbar^2(1/I_z - 1/I_x)$ [1,4]. This approximation would result in the inactive rotational energy being adiabatic (i.e. a function of J only); a result that certainly simplifies the expression for the microscopic rate constant with K active, $k^a(E, J)$. However, for this approximation to be valid, E_K must be small compared to the rotational energy in the x, y plane [1]. This latter quantity, for a symmetric top, is

$$E_{r,xy} = \frac{L_x^2 + L_y^2}{2I_x} = \frac{[J(J+1) - K^2]\hbar^2}{2I_x}. \quad (7)$$

However, E_K may be greater than $E_{r,xy}$ for large J s (≈ 25) and some values of K , so that this is not, in general, a good approximation; consequently, one

needs to treat the active and inactive portions of the rotational energy as explicit functions of K , and J and K , so that the active energies become

$$E_a^* = E_{\text{int}}^* + E_K^* = E - E_0 - E_r^*(J, K) + K^2\hbar^2/2I_z^* \\ = E - E_0 - [J(J+1) - K^2]\hbar^2/2I_x^*. \quad (8a)$$

and

$$E_a = E_{\text{int}} + E_K = E - [J(J+1) - K^2]\hbar^2/2I_x. \quad (8b)$$

Since the square of the angular momentum along the z axis is

$$L_z^2 = K^2\hbar^2,$$

then conservation of angular momentum requires that the square of the angular momentum about the two equal axes (the x, y plane) be

$$\langle L_x^2 + L_y^2 \rangle = [J(J+1) - K^2]\hbar^2.$$

The microscopic rate constant, $k^a(E, J)$, is given by

$$k^a(E, J) = \frac{r^* G_{i,K}^*(E_a^*)}{h N_{i,K}(E_a)}, \quad (9)$$

in which $G_{i,K}^*$ and $N_{i,K}$ are convolutions of the internal levels with the K levels, and are given by

$$G_{i,K}^*(E_a^*) = \sum_K G_i^*(E_a^* - E_K^*) N_K(E_K^*),$$

$$K = -J \text{ to } +J, \quad (10a)$$

$$N_{i,K}(E_a) = \sum_K N_i(E_a - E_K) N_K(E_K),$$

$$K = -J \text{ to } +J. \quad (10b)$$

Since the argument of G_i^* is limited to positive values, then Eq. (10a) reduces to

$$G_{i,K}^*(E_a^*) = \sum_K^+ G_i^*(E - E_0 - E_r^*(J, K)), \quad (11a)$$

and by similar reasoning, one transforms Eq. (10b) to

$$N_{i,K}(E_a) = \sum_K^+ N_i(E - E_r(J, K)), \quad (11b)$$

where the symbol, Σ^+ , is introduced to emphasize that the summation is over all values of K corresponding to positive values of $E - E_r(J, K)$ rather than positive values of $E - E_0 - E_r^*(J, K)$, as in Eq. (5).

Substituting Eqs. (11) into Eq. (9), one obtains

$$k^a(E, J) = \frac{r^* \sum_K^+ G_i^+(E - E_0 - E_r^+(J, K))}{h \sum_K^+ N_i(E - E_r(J, K))},$$

$$K = -J \text{ to } +J. \quad (12)$$

From the definition of G_i^+ and N_i for negative arguments (namely $G_i^+(a) = N_i(a) = 0$; $a < 0$) one can readily see that Eq. (12) is the same as that of Miller [7] and Zhu and Hase [12] and that, at high E and/or low J , Eq. (12) will equal Eq. (5) too. However, at energies close to threshold, Eq. (12) will generally be smaller than Eq. (5), and that a large value for E_0 will accentuate this difference.

The unimolecular rate constant with K active becomes

$$k_{\text{uni}}^a = \int_{E_0}^{\infty} \sum_J k^a(E, J) \frac{w}{w + k^a(E, J)} \times \frac{(2J+1) \sum_K^+ N_i(E - E_r(J, K))}{Q_r Q_{\text{int}}} e^{-E/kT} dE. \quad (13)$$

Similar to Eq. (6), the summation over J in Eq. (13) should be over those values of J for which $k^a(E, J) > 0$.

4. Results

It is evident from the defining relationships for $\langle k(E, J) \rangle_R$ and $k^a(E, J)$ that the high-pressure limits of Eqs. (6) and (13) will be identical, however, not so for the low-pressure limits since

$$\sum_K^+ N_i(E - E_r(J, K)) \geq \sum_K^+ N_i(E - E_r(J, K)).$$

Consequently, we can illustrate the importance of the three-dimensional model by comparing the low-pressure limits of Eqs. (6) and (13) with those obtained in the two-dimensional model. In this initial work we have chosen two reactions; one with a prolate top ($\text{C}_2\text{H}_6 \rightarrow 2\text{CH}_3$), and the other with an oblate top ($\text{HCCl}_3 \rightarrow \text{HCl} + \text{CCl}_2$).

The vibrational assignment for ethane and moments of inertia for ethane and its transition state were taken from Stewart et al. [14]. The vibrational assignments are nominally equivalent to those of Chao et al. [15] and Waage and Rabinovitch [16]. The hindered internal rotation in ethane was treated

as a torsional frequency. E_0 was 87.78 kcal/mol [14]. One can calculate a value 0.3 kcal/mol larger from the TRC tables [17,18] which, if used, would cause the k_0 for ethane (Table 1) to be changed by $e^{-151/T}$.

The frequencies and moments of inertia for HCCl_3 were taken from Rodgers et al. [19], and the moments of inertia of the transition state were estimated as $I_x^+/I_x = 1.3$ and $I_z^+/I_z = 1.0$. The value, ${}_rH_0^0 = 55.9$ kcal/mol, was calculated from the TRC tables [20,21] and used to approximate E_0 for the reaction, $\text{CHCl}_3 \rightarrow \text{CCl}_2 + \text{HCl}$, this value is consistent with the experimental data and a very low barrier for the reverse reaction [22].

The densities were calculated at 10 cm^{-1} intervals with the Beyer–Swinehart algorithm [23] and then linearly interpolated. The frequency assignments were, therefore, judiciously rounded to 10 cm^{-1} . The collision frequencies (w) were calculated from Lennard-Jones 6–12 potential parameters in the strong collision limit with argon as the collider [24].

The results obtained for ethane (a prolate top) are summarized in Table 1, and were obtained by summation over K , J , and E . Those for chloroform (an oblate top) are given in Table 2. Since the rotational levels were very dense for chloroform, these results were obtained by numerical integration over J ; the trapezoid rule was used with $J=20$. This reduced computational time by a factor of at least 20 and gave excellent agreement with one rate constant done by direct count.

Table 1

Comparison of the low-pressure limit of the strong collision bimolecular rate constants, $k_0/[M]$, as calculated by the two- and three-dimensional models^a for the reaction: $\text{C}_2\text{H}_6 + \text{Ar} \rightarrow 2\text{CH}_3 + \text{Ar}$

Temp. (K)	$k_0/[M] \text{ (cm}^3 \text{ mol}^{-1} \text{ s}^{-1}\text{)}$			
	K rotor active		K rotor adiabatic	
	3D	2D	3D	2D
800	4.56	2.63	0.927	1.27
1000	1.05×10^5	6.86×10^4	2.78×10^4	3.68×10^4
1200	6.04×10^7	4.31×10^7	1.96×10^7	2.53×10^7

^a See text for details of the calculation.

Table 2

Comparison of the low-pressure limit of the strong collision bimolecular rate constants, $k_0/[M]$, as calculated by the two- and three-dimensional models^a for the reaction: $\text{CHCl}_3 + \text{Ar} \rightarrow 2:\text{CCl}_2 + \text{HCl} + \text{Ar}$

Temp. (K)	$k_0/[M]$ ($\text{cm}^3 \text{mol}^{-1} \text{s}^{-1}$)			
	<i>K</i> rotor active		<i>K</i> rotor adiabatic	
	3D	2D	3D	2D
800	1.10×10^6	2.68×10^5	1.38×10^5	1.60×10^5
1000	4.57×10^8	1.35×10^8	7.78×10^7	8.94×10^7
1200	2.03×10^{10}	6.93×10^9	4.35×10^9	5.02×10^9

^a See text for details of the calculation.

5. Conclusions

The results summarized in Tables 1 and 2 lead to two conclusions, the most important of which is that, if the *K* rotor is active (and this is the usual assumption), then conservation of angular momentum in all three dimensions leads to a significant increase in the low-pressure rate constant when compared with the two-dimensional model. For the examples shown here over the temperature range of 1200 to 800 K, this increase varies from 40% to 70% for ethane and from 190% to 310% for chloroform. For each reaction, the activation energy of the low-pressure rate constant is 1 to 2 kcal/mol less than that for the two-dimensional model. Further, Eq. (7) et seq. clearly shows that for this model the orbital angular momentum is not conserved. The rotational energy associated with this motion serves as an energy reservoir for the active energy, similar to the action of a flywheel.

When the *K* rotor is adiabatic, the three-dimensional model yields lower values for k_0 than the two-dimensional model. The decrease is in the range of 15% to 30% and the activation energies are nominally the same, so that the two-dimensional model is a reasonable approximation in this case.

Finally, it is interesting to note that the collisional efficiency determined from the three-dimensional model with the *K* rotor active is more self-consistent than those from the two-dimensional models. Using experimental data for $k_0/[\text{Ar}]$ at 1200 K [14,22], and defining the collisional efficiency, β , as $k_0(\text{exp})/k_0^{\text{sc}}(\text{theory})$, one obtains $\beta=0.048$, whereas the two-dimensional models and the three-dimen-

sional model with *K* adiabatic lead to values differing from one another by 50% to 100%. Furthermore, $\beta \approx 0.05$ at 1200 K is not unreasonable in view of energy transfer experiments [1].

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